```sql
USE hotel_analysis;
SELECT *FROM hotel
LIMIT 10;
SELECT ID, n_adults, n_children, room_type, year
FROM hotel
WHERE n_adults >= 2;
SELECT ID, lead_time, year, month
FROM hotel
ORDER BY lead_time DESC
LIMIT 10;
SELECT room_type,
       AVG(avg_room_price) AS average_price
FROM hotel
GROUP BY room_type;

SELECT SUM(avg_room_price) AS total_room_price
FROM hotel;

SELECT AVG(avg_room_price) AS overall_average_price
FROM hotel;

SELECT COUNT(*) AS total_bookings
FROM hotel;

DROP TABLE IF EXISTS room_types;

CREATE TABLE room_types (
    room_type VARCHAR(50),
    room_description VARCHAR(100)
);
INSERT INTO room_types (room_type, room_description)
VALUES
('Room_Type 1', 'Standard Room'),
('Room_Type 2', 'Deluxe Room'),
('Room_Type 3', 'Executive Room'),
('Room_Type 4', 'Family Room'),
('Room_Type 5', 'Premium Room'),
('Room_Type 6', 'Suite Room'),
('Room_Type 7', 'Luxury Room');
SELECT h.ID,
       h.room_type,
       r.room_description,
       h.avg_room_price
FROM hotel h
INNER JOIN room_types r
ON h.room_type = r.room_type
LIMIT 20;
SELECT h.ID,
       h.room_type,
       r.room_description
FROM hotel h
LEFT JOIN room_types r
ON h.room_type = r.room_type
LIMIT 20;
SELECT h.ID,
       h.room_type,
       r.room_description
FROM hotel h
RIGHT JOIN room_types r
ON h.room_type = r.room_type;
SELECT ID,
       room_type,
       avg_room_price
```

```
FROM hotel
WHERE avg_room_price >
      (
          SELECT AVG(avg_room_price)
          FROM hotel
      );
DROP VIEW IF EXISTS room_price_analysis;
CREATE VIEW room_price_analysis AS
SELECT room_type,
       COUNT(*) AS total_bookings,
       AVG(avg_room_price) AS average_price
FROM hotel
GROUP BY room_type;
SELECT *
FROM room_price_analysis;

CREATE INDEX idx_lead_time
ON hotel(lead_time);

SHOW INDEX FROM hotel;
```

hotel_booking_analysis.sql